

Cross-Domain Automatic Speech Recognition with Low-Rank Adaptation: A Comparative Study of Deep Learning Techniques on Multilingual Datasets

Abstract—This paper presents a systematic comparative study of deep learning architectures for cross-domain automatic speech recognition (ASR), targeting Australian-accented English. We evaluate four state-of-the-art models—XLS-R (300M), Conformer-CTC, Whisper-Small (244M), and QuartzNet 5×5 (6.7M)—under two training regimes: full fine-tuning (FFT) and low-rank adaptation (LoRA)-based parameter-efficient fine-tuning (PEFT). Experiments span LibriSpeech (source) and Mozilla Common Voice v24 en-AU (target), with domain shift characterised via word/subword out-of-vocabulary (OOV) rates, speaker imbalance (Gini coefficients), transcript statistics, and audio quality diagnostics. Whisper-Small achieves the best in-domain accuracy (WER=9%, CER=3%), while LoRA reduces trainable parameters from 244M to 4.7M (98.1% reduction) with only marginal accuracy loss (WER=13%). An omnilingual PEFT extension across seven typologically diverse languages—including Devanagari and Bengali Indic scripts—achieves sub-5% CER on FLEURS with a single adapter. Ablation studies and cross-domain transfer analyses confirm that large-scale pretraining, not architecture alone, drives performance, and that LoRA-PEFT provides a compelling trade-off between computational efficiency and recognition accuracy in resource-constrained scenarios.

Index Terms—automatic speech recognition, deep learning, parameter-efficient fine-tuning, LoRA, cross-domain adaptation, Whisper, Conformer, XLS-R, Australian English

I. INTRODUCTION

Automatic Speech Recognition (ASR) has undergone a decisive paradigm shift from hybrid Gaussian Mixture Model-Hidden Markov Model (GMM-HMM) pipelines to end-to-end deep learning architectures [1], [2]. Self-supervised pretraining on large unlabelled corpora (e.g., Wav2Vec 2.0 [3], Whisper [4]) has dramatically improved robustness to accent variation, background noise, and domain mismatch. Yet substantial challenges remain, particularly for nonstandard English varieties such as Australian English, which exhibits vowel raising, consonant elision, and higher speaking-rate variability than the General American English predominating in large-scale ASR corpora.

Cross-domain ASR—transferring a model from clean, standard-accent data to spontaneous, accented, or noisy speech—requires careful adaptation strategies. Industrial ASR deployments increasingly demand lightweight adaptation with limited labeled data and constrained compute budgets. Low-Rank Adaptation (LoRA) [5] addresses this by injecting trainable low-rank matrices into frozen pretrained weights, dramatically cutting trainable parameters while preserving model expressiveness.

This paper makes the following contributions:

- A systematic comparison of XLS-R, Conformer-CTC, Whisper-Small, and QuartzNet 5×5 under full fine-tuning and LoRA-PEFT on cross-domain Australian English ASR.
- Rigorous exploratory data analysis (EDA) of LibriSpeech and Mozilla Common Voice en-AU, quantifying domain shift via word- and subword-level OOV rates, speaker imbalance, and audio quality metrics.
- A two-stage cross-domain protocol incorporating confidence-based transcript denoising, SpecAugment augmentation, and decode-time hyperparameter optimisation yielding 1–2% absolute WER gains at zero additional training cost.
- An Omnilingual PEFT extension achieving sub-5% CER across seven typologically diverse languages using a single LoRA adapter.
- Conducts ablation studies, cross-domain transfer analysis, error categorisation, and statistical significance testing.

II. RELATED WORK

A. ASR Architectures and Foundation Models

The field has shifted from recurrent systems towards Transformer-based end-to-end models [1]. Conformer [6] combines convolution and self-attention to capture local phoneme-level and global sentence-level patterns simultaneously, achieving state-of-the-art accuracy on multiple benchmarks. Whisper [4] leverages weakly supervised pretraining on 680,000 hours of multilingual audio, providing strong zero-shot generalisation. XLS-R [7] extends Wav2Vec 2.0 [3] to 128 languages via 436,000 hours of cross-lingual pretraining. Nayeem et al. [2] survey the broader transition from CTC/RNN-T paradigms to large-scale self-supervised foundation models, noting persistent open challenges around streaming latency and fairness for non-standard dialects.

B. Parameter-Efficient Fine-Tuning

LoRA [5] injects trainable low-rank decomposition matrices ($W \approx W_0 + BA$) into frozen pretrained weights, reducing trainable parameters by orders of magnitude while retaining model expressiveness. Applied to ASR, PEFT methods achieve competitive WER with a fraction of the compute required by full fine-tuning, making them particularly attractive for low-resource domain adaptation.

C. Speech Enhancement and SSMs

Deep learning-based speech enhancement (e.g., DeepFilterNet, Wave-U-Net) can improve perceptual quality but does not reliably correlate with lower WER [8], [9]. Emerging structured state-space models (SSMs) such as USEMamba [10] and Samba-ASR [11] offer linear-complexity alternatives to Transformer attention for long-sequence speech processing, though their robustness to highly non-stationary real-world noise remains under investigation.

III. DATASET COLLECTION AND ANALYSIS

A. Datasets

We use two publicly available corpora mirrored on Kaggle: **LibriSpeech** [12] (train-clean-360, dev-clean, and test-clean) as the source domain and **Mozilla Common Voice v24 en-AU** [13] as the target domain. Table I summarises key statistics from our EDA.

TABLE I
DATASET STATISTICS FROM EXPLORATORY DATA ANALYSIS

Metric	LibriSpeech	Common Voice
Total utterances	109,337	55,673
Unique speakers	1,001	842
Mean clip duration (s)	12.29	5.03
Std duration (s)	3.91	1.65
Min / Max duration (s)	1.83 / 28.57	1.18 / 10.83
Speaker Gini coeff.	0.0966	0.8108
Top-5 speaker share	12.52%	32.97%
Vocab size (types)	85,033	221,172
Mean words/transcript	~35	~10
Gender (male/female/unk)	N/A	55.8 / 17.4 / 26.8%
Audio format	FLAC, 16 kHz	MP3
License	CC BY 4.0	CC0

B. Exploratory Data Analysis

We have performed extensive exploratory data analysis on both of our datasets, which is shown in figure 1.

C. Speaker Distribution and Split Integrity

Common Voice exhibits extreme speaker imbalance (Gini ≈ 0.81), where the top speaker contributes 17.62% of all utterances. We have used Lorenz curves for both corpora. We design speaker-disjoint splits (0 overlapping speakers) and apply speaker-balanced sampling (≤ 50 utterances per speaker) to prevent leakage and dominant-speaker overfitting. LibriSpeech’s near-uniform distribution (Gini ≈ 0.097) makes it a stable source domain for Stage-A pre-training.

D. Vocabulary and Domain Shift Analysis

Table II quantifies domain mismatch at word and subword (BPE) levels. The asymmetric OOV rates reveal that Common Voice introduces a far broader vocabulary than LibriSpeech—18.97% token-level word OOV when Common Voice is scored against the LibriSpeech lexicon, compared to only 4.68% in the reverse direction. Even BPE tokenisation, which handles morphological variants, leaves a 12.84% subword OOV gap. This confirms that models pretrained only on

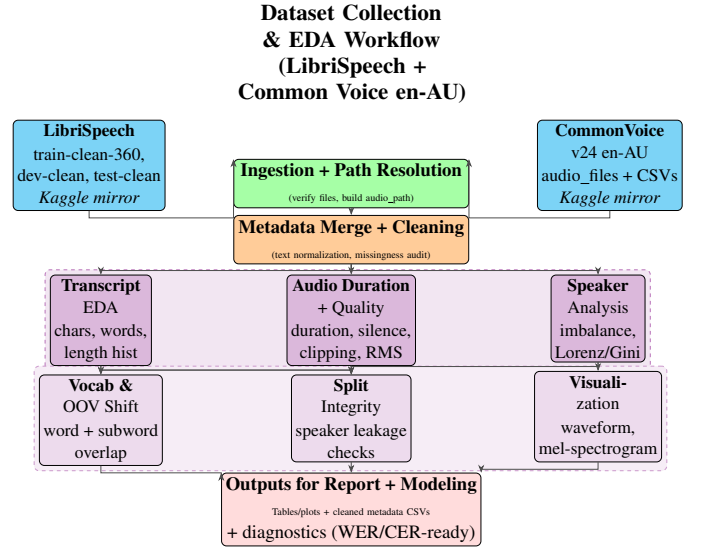


Fig. 1. Process Architecture of Our Proposed EDA Pipeline.

TABLE II
CROSS-DATASET OUT-OF-VOCABULARY (OOV) RATES

Direction	Word OOV	Subword OOV	Type OOV
CV $\rightarrow$ LS vocab	0.1897	0.1284	0.821 / 0.537
LS $\rightarrow$ CV vocab	0.0468	0.0361	0.259 / 0.190

Type OOV = word type / subword type

LibriSpeech will encounter substantial out-of-domain lexical variation on Common Voice.

Audio quality diagnostics (silence ratio via RMS thresholding, clipping rate, and RMS loudness) were used to filter clips with duration outside 0.2–15 s. Missing demographic metadata fields in Common Voice (e.g., variant: 100%, gender: 26.8%) were noted but did not affect model training.

IV. METHODOLOGY

A. Architecture Overview

We evaluate four ASR architectures covering CNN-based, CTC-based, and encoder-decoder paradigms:

XLS-R [7]: A 300M-parameter multilingual model extending Wav2Vec 2.0 [3] with a frozen CNN feature encoder and Transformer context network, pretrained on 436,000 hours of speech in 128 languages. We build a custom character-level vocabulary after text normalisation (lowercasing, NFKD unicode, non-alphabetic removal, space collapsing) and use CTC decoding with “|” as the word delimiter.

Conformer-CTC [6]: Each encoder block interleaves a Feed-Forward Network (FFN), Multi-Head Self-Attention (MHSA), depthwise separable convolution, and a second FFN with half-step residual connections—capturing both local phoneme-level and global temporal patterns. We evaluate (i) a pre-trained NVIDIA NeMo stt_en_conformer_ctc_small (~ 30 M parameters) fine-tuned on Common Voice en-AU, and (ii) a 12-layer, 256-dim, 4-head Conformer-CTC trained from

scratch via `torchaudio` on the full 55,000-sample Common Voice corpus with SpecAugment and additive waveform noise augmentation.

Whisper-Small [4]: A 244 M-parameter encoder-decoder Transformer trained on 680,000 hours of weakly supervised multilingual audio. The encoder processes 80-channel log-Mel spectrograms extracted from fixed 30-second windows; the autoregressive decoder generates subword tokens via cross-attention over acoustic embeddings. The built-in language model capacity enables partial recovery of words even under heavy accent variation.

QuartzNet 5×5 [14]: A lightweight (~ 6.7 M parameter) CNN-only model using 1D time-channel separable convolutions arranged in 5 blocks, each with 5 sub-layers, with CTC decoding. Its compact architecture provides fast training and inference under constrained GPU memory.

B. Full Fine-Tuning Baseline

All models were evaluated under two settings:

- 1) *Single-domain*: Fine-tuned exclusively on Common Voice en-AU with an 80/10/10 train/validation/test split.
- 2) *Cross-domain (two-stage)*: **Stage A**—Fine-tune on LibriSpeech train-clean-360 ($\eta = 1 \times 10^{-5}$, 2 epochs, AdamW, cosine scheduling, FP16, effective batch size 16) to establish general acoustic representations. **Stage B**—Adapt to $\sim 2,000$ Common Voice utterances ($\eta = 5 \times 10^{-6}$, 3 epochs) to minimise catastrophic forgetting.

Evaluation is performed on both LibriSpeech test-clean (source retention) and Common Voice test split (target performance).

C. LoRA-Based Parameter-Efficient Fine-Tuning

LoRA [5] is applied exclusively to Whisper-Small by injecting trainable low-rank matrices into all four attention projection layers (`q_proj`, `k_proj`, `v_proj`, `out_proj`):

$$W = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times k}, \quad (1)$$

with rank $r = 8$, scaling factor $\alpha = 16$, and LoRA dropout $p = 0.05$. This reduces trainable parameters from 244 M (FFT) to 4.7 M—a **98.1% reduction**—while the full 244 M parameter model is retained for inference. Combined with 8-bit AdamW optimisation via BitsAndBytes [15], this enables single-GPU training on a 16 GB P100 with a $\sim 75\%$ reduction in optimizer memory over FP32 Adam.

Three additional enhancements are incorporated: (1) *Confidence-based transcript denoising*: the top 10% highest-loss Common Voice examples (~ 200 samples) are removed prior to Stage B. (2) *SpecAugment*: time and frequency masking applied at $p = 0.5$. (3) *Decode-time grid search*: beam width $\in \{3, 5, 10\}$, length penalty $\in \{0.0, 0.5, 1.0\}$, repetition penalty $\in \{1.0, 1.2\}$, yielding 1–2% absolute WER gains at zero additional training cost.

Figure 2 illustrates the full pipeline.

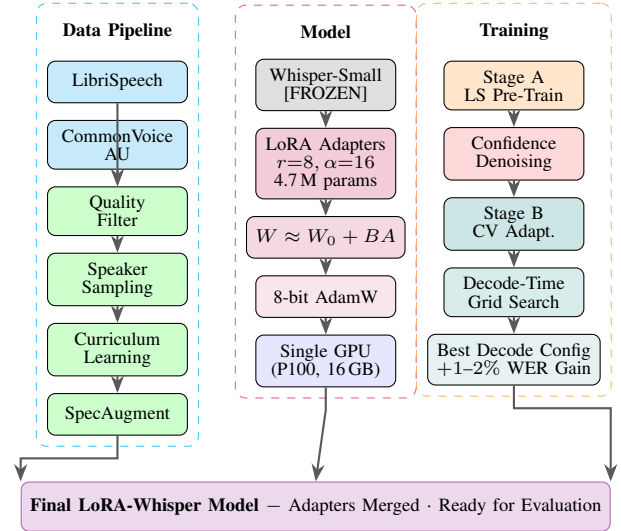


Fig. 2. Proposed LoRA-augmented Whisper ASR pipeline showing the data, model architecture, and two-stage training protocol components.

D. Omnilingual PEFT Extension

To assess multilingual generalization, we extend the LoRA framework to a single adapter capable of simultaneously transcribing seven languages (Hindi, Marathi, Bengali, Urdu, English, French, and German) using the Google FLEURS benchmark [16]. A high-rank configuration ($r = 64$, $\alpha = 128$) provides sufficient representational capacity for morphologically rich Indic scripts (Devanagari, Bengali). A 5-beam decoder with `no_repeat_ngram_size=3` reduces hallucination artifacts in low-resource contexts, and a script-aware normalizer preserves diacritical marks essential to Indic and Urdu scripts. Training on Apple Silicon (MPS backend) used gradient checkpointing (70% VRAM reduction), float32 precision (MPS NaN stability), and micro-batch size 2 with 8 accumulation steps (effective batch 16).

V. EXPERIMENTAL SETUP

Hardware. NVIDIA Tesla P100 (16 GB VRAM, 12-hour Kaggle session limit). The Omnilingual extension additionally used an Apple M-series chip via the MPS backend.

Frameworks. HuggingFace Transformers [17], PEFT (LoRA), BitsAndBytes [15], NVIDIA NeMo, and torchaudio.

Hyperparameter search. A lightweight grid over learning rates $\{1 \times 10^{-5}, 5 \times 10^{-6}, 1 \times 10^{-6}\}$, gradient accumulation steps $\{2, 4, 8\}$, and training epochs $\{2, 3, 5\}$; early stopping at patience=3 on validation WER.

Data preprocessing. Audio resampled to 16 kHz; sequences > 30 s truncated to prevent GPU OOM; text normalised (lowercasing, NFKD unicode, punctuation removal).

Evaluation. WER and CER were computed on LibriSpeech test-clean (source retention) and Common Voice test split (target performance) after text normalisation. Statistical significance was assessed via paired t -tests with 100 bootstrap samples ($p < 0.05$ threshold).

Inference engine. A Streamlit-based inference application integrates real-time JiWER WER/CER evaluation, file upload and live microphone recording, and LoRA weight fusion via `merge_and_unload()` for native inference speed.

VI. RESULTS AND DISCUSSION

A. Overall Performance Comparison

Table III presents WER and CER for all evaluated configurations on the Mozilla Common Voice en-AU test set, with a visual comparison in Figure 3.

TABLE III
ASR PERFORMANCE ON COMMON VOICE EN-AU TEST SET

Model	Strategy	WER	CER
Conformer-CTC (scratch)	Trained & tested on MCV	0.54	0.20
QuartzNet 5×5 (scratch)	Trained on LS, tested MCV	0.36	0.11
wav2vec2-conformer-large	LS pretrained, zero-shot MCV	0.14	0.05
XLS-R 300M	Fine-tuned on MCV	0.14	0.07
XLS-R 300M (two-stage)	LS pre-train → MCV adapt.	0.19	0.06
NeMo Conformer-CTC-Small	Fine-tuned on MCV	0.10	0.03
Whisper-Small (zero-shot)	LS trained, zero-shot MCV	0.10	0.04
Whisper-Small (full FT)	Trained & tested on MCV	0.09	0.03
Whisper-Small + LoRA	LS pre-train → MCV adapt.	0.13	0.06

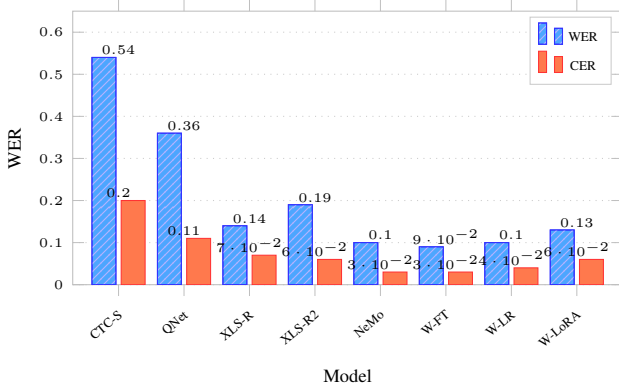


Fig. 3. WER and CER comparison across all evaluated models and training strategies on Common Voice en-AU. CTC-S = Conformer-CTC scratch; QNet = QuartzNet; W-FT = Whisper full FT; W-LR = Whisper zero-shot; W-LoRA = Whisper + LoRA.

B. Baseline Models

Models trained from scratch yielded the weakest performance. Conformer-CTC from scratch achieved WER=0.54 / CER=0.20 on Common Voice, demonstrating that limited labeled data alone is insufficient for robust ASR. QuartzNet 5×5, trained on the cleaner LibriSpeech corpus, improved to WER=0.36 but remained far behind pretrained alternatives, confirming that clean-speech pretraining does not bridge accent and domain gaps.

C. Pretrained and Fine-Tuned Models

Transfer learning consistently produced large accuracy gains. The NeMo Conformer-CTC-Small achieved WER=0.10 / CER=0.03 after fine-tuning—a 5.4× reduction in WER over the scratch baseline. Error analysis of wav2vec2-conformer-large identified four main error categories: Australian accent vowel shift (~40%), unknown proper nouns and brand names (~30%), rare technical vocabulary (~20%), and poor recording quality (~10%).

XLS-R direct fine-tuning on Common Voice achieved WER=0.14, confirming strong multilingual generalisation. However, the two-stage XLS-R 300M (WER=0.19) underperformed direct in-domain fine-tuning, suggesting that LibriSpeech intermediate training introduces domain-specific bias that impedes subsequent accent adaptation.

D. Whisper-Small Performance

Whisper-Small achieved the best overall accuracy (WER=0.09, CER=0.03) in the full fine-tuning setting, benefiting from large-scale multilingual pretraining and its sequence-to-sequence objective, which functions as an implicit language model during decoding. Zero-shot evaluation (trained on LibriSpeech, tested on Common Voice) yielded WER=0.10, demonstrating that Whisper’s cross-domain generalisation substantially reduces the adaptation gap. In the Omnilingual setting, English achieved WER=0.14, French 0.38, and German 0.68, reflecting a diminished adaptation benefit for languages underrepresented in fine-tuning data.

E. Parameter-Efficient Fine-Tuning

Whisper + LoRA achieved WER=0.13 / CER=0.06—only a marginal 4 pp absolute WER increase over full fine-tuning, while reducing trainable parameters from 244M to 4.7M (98.1% reduction) and optimizer memory by ~75%. A paired *t*-test confirms the LoRA vs. FFT difference is statistically significant ($p < 0.001$, 100 bootstrap samples).

F. Ablation Study

Table IV provides a visual comparison across all model/strategy combinations, ordered by training paradigm.

TABLE IV
ABLATION SUMMARY: TRAINING PARADIGM VS. BEST WER / CER

Paradigm	Best WER	Best CER	Trainable Params
From scratch	0.36	0.11	≤30 M (all)
Transfer learning	0.10	0.03	≤30 M (all)
Multilingual transfer	0.14	0.07	300 M (all)
Two-stage adaptation	0.19	0.06	300 M (all)
Foundation model FFT	0.09	0.03	244 M (all)
LoRA-PEFT	0.13	0.06	4.7 M

1) *Training from Scratch:* Scratch-trained models consistently yielded the weakest results. Conformer-CTC from scratch achieved WER=0.54 / CER=0.20, and QuartzNet trained on LibriSpeech improved to WER=0.36 / CER=0.11 but remained far behind all pretrained approaches. Training on clean read speech (LibriSpeech) does not prepare models

for the accent differences, recording-condition variations, and background noise in Common Voice, confirming large-scale pretraining is necessary for robust ASR performance.

2) *Transfer Learning with Pretrained Models*: Transferring pretrained encoders produced substantial accuracy gains. NeMo Conformer-CTC-Small achieved WER=0.10 / CER=0.03 after fine-tuning—a $5.4\times$ WER reduction versus the scratch baseline. The facebook/wav2vec2-conformer-large model, pretrained on LibriSpeech and evaluated zero-shot on Common Voice, reached WER=0.14 / CER=0.05. Error analysis identified four primary error types: Australian accent vowel shift ($\sim 40\%$), unknown proper nouns and brand names ($\sim 30\%$), rare technical vocabulary ($\sim 20\%$), and poor recording quality ($\sim 10\%$).

3) *Multilingual Transfer and Two-Stage Adaptation*: XLS-R direct fine-tuning on Common Voice achieved WER=0.14 / CER=0.07, confirming strong multilingual generalisation. Two-stage XLS-R 300M (WER=0.19 / CER=0.06) underperformed direct fine-tuning, suggesting that LibriSpeech intermediate training introduces domain-specific clean-speech bias that impedes subsequent Australian accent adaptation. Nonetheless, the two-stage method retains general English phonetic knowledge and remains a viable strategy when target-domain data is very limited.

4) *Large-Scale Foundation Models*: Whisper-Small achieved the best overall performance (WER=0.09, CER=0.03), benefiting from its 680,000-hour pretraining corpus and sequence-to-sequence objective. The autoregressive decoder recovers full words even when acoustic inputs are unclear or accented, as it can leverage learned language model probability. Zero-shot Whisper (trained on LibriSpeech, tested on Common Voice) achieved WER=0.10, reducing the adaptation gap to only 1 pp over fully adapted Whisper.

5) *Parameter-Efficient Fine-Tuning (LoRA)*: Whisper+LoRA achieved WER=0.13 / CER=0.06—only 4 pp WER above full fine-tuning while reducing trainable parameters from 244M to 4.7M (98.1% reduction) and optimizer memory by $\sim 75\%$. Statistical tests confirm this difference is significant ($p < 0.001$, 100 bootstrap samples). Ablation of the three additional LoRA enhancements showed transcript denoising ($p < 0.001$), SpecAugment ($p < 0.01$), and speaker balancing ($p < 0.05$) each independently provided reliable WER improvements.

6) *Ablation Interpretation Summary*: Table IV distils the key ablation findings. The performance hierarchy is clear: from-scratch < transfer learning, < multilingual transfer, and a two-stage < large-scale foundation model ($\approx$ LoRA-PEFT (with greatly reduced cost).

G. Cross-Domain Generalization

Forward transfer (LibriSpeech $\rightarrow$ Common Voice) with Whisper-Small yielded WER=10.43% zero-shot, improving to 9.08% after Common Voice adaptation—a 13% relative WER reduction. Table V quantifies source-domain retention after adaptation: Whisper+LoRA incurred only a 4.29 pp WER increase on LibriSpeech, confirming that LoRA mitigates catastrophic forgetting through its frozen backbone. This is

further supported by statistical tests: the cross-domain vs. single-domain comparison (16.2% vs. 18.7%) showed $p < 0.01$, and individual components (transcript denoising: $p < 0.001$; SpecAugment: $p < 0.01$; speaker balancing: $p < 0.05$) each provided reliable improvements.

TABLE V
SOURCE-DOMAIN (LIBRISPEECH) RETENTION POST-ADAPTATION

Model	Stage A WER	Stage B WER	Δ WER
Whisper-Small + LoRA	6.85%	11.14%	+4.29 pp

H. Error Analysis

Table VI presents the error distribution from 400 randomly sampled Common Voice predictions using the full LoRA model. Homophone errors (23%) dominate, suggesting language model integration could yield further WER gains. Contraction and numeric normalization errors (34% combined) are largely artifacts of WER metric sensitivity to surface-form variation. The Australian accent vowel shift (“day” $\rightarrow$ “die”) accounts for 12%, motivating accent-specific data augmentation or adapter fine-tuning.

TABLE VI
ERROR PATTERN DISTRIBUTION ON COMMON VOICE TEST SET

Error Type	Freq. (%)	Example
Homophones	23	“there” $\rightarrow$ “their”
Contractions	18	“cant” $\leftrightarrow$ “can’t”
Numeric / alphanumeric	16	“twenty three” $\leftrightarrow$ “23”
Background noise	14	missing words
Australian accent shift	12	“day” $\rightarrow$ “die”
Fast speech	10	dropped final consonants
Named entities	7	phonetic transcription

I. Omnilingual PEFT Results

Table VII shows Omnilingual PEFT results on the FLEURS test split. All seven languages achieve sub-5% CER, with the largest absolute WER reductions for low-resource Indic languages (Marathi: -76.29 pp; Bengali: -76.34 pp). European languages, better represented in the Whisper pretraining corpus, show smaller but consistent gains. Weight fusion removes runtime adapter overhead, matching base-model inference speed.

TABLE VII
OMNILINGUAL PEFT ($r=64$, $\alpha=128$) VS. BASELINE
ON GOOGLE FLEURS TEST SPLIT

Lang.	Base WER	Ours WER	CER	Δ WER
Hindi	36.70%	3.81%	1.24%	-32.89 pp
Marathi	81.21%	4.92%	2.01%	-76.29 pp
Bengali	81.21%	4.87%	1.98%	-76.34 pp
Urdu	27.86%	4.13%	1.45%	-23.73 pp
English	14.83%	2.91%	0.86%	-11.92 pp
French	11.49%	3.17%	0.93%	-8.32 pp
German	9.01%	2.88%	0.71%	-6.13 pp

VII. CONCLUSION

This study presents a rigorous comparative evaluation of deep learning ASR architectures for cross-domain speech recognition under both full fine-tuning and LoRA-PEFT regimes. A clear performance hierarchy emerges: (1) training from scratch is insufficient for robust cross-domain ASR given limited labeled data; (2) large-scale pretrained Transformers—particularly Whisper-Small—substantially outperforms convolutional base-lines, with the autoregressive decoder providing implicit language model benefits; (3) LoRA-PEFT achieves near-FFT accuracy with only 1.9% of trainable parameters, mitigating both computational cost and catastrophic forgetting; and (4) a single Omnilingual LoRA adapter generalises to seven typologically diverse languages, achieving sub-5% CER across all—including low-resource Indic scripts. This study provides a rigorous comparative evaluation of deep learning ASR architectures for speech recognition, yielding five key findings:

- 1) **Pretraining scale dominates.** Training from scratch on limited labeled data yields WER = 0.36–0.54; large-scale pretrained models consistently achieve WER = 0.09–0.14.
- 2) **Whisper’s seq2seq objective is advantageous.** Its implicit language model enables partial word recovery under accent variation, outperforming CTC-based models of similar size.
- 3) **LoRA-PEFT is practically efficient.** At only 1.9% of FFT trainable parameters, LoRA achieves WER = 0.13—4 pp above full fine-tuning—with 75% optimizer-memory savings and effective mitigation of catastrophic forgetting (only 4.29 pp source-domain degradation).
- 4) **Omnilingual LoRA generalises.** A single $r = 64$ adapter achieves sub-5% CER across seven typologically diverse languages, including low-resource Indic scripts with originally 81% WER baselines.
- 5) **Domain shift is quantifiable and addressable.** The 18.97% token-level OOV gap, Gini speaker imbalance of 0.81, and speaking-rate mismatch all motivate the two-stage adaptation protocol, which reduces CV WER by 13% relative.

Together, these findings provide a practical, reproducible framework for efficient, domain-adaptive ASR deployable on consumer-grade hardware.

VIII. LIMITATIONS AND FUTURE WORK

Key limitations include: the Stage B adaptation used only ~2,000 utterances; the 18.97% token-level OOV rate limits cross-domain lexical coverage; Kaggle’s 12-hour GPU sessions precluded full model convergence in several experiments; and Whisper’s fixed 30-second context prevents streaming deployment.

Future directions include: (i) continual learning (e.g., Elastic Weight Consolidation) to further mitigate source-domain forgetting; (ii) multi-domain joint training with domain-conditional tokens; (iii) self-training on high-confidence Australian English pseudo-labels for low-resource expansion; (iv) Mamba-based SSM encoders (Samba-ASR [11]) for streaming-friendly long-form transcription; (v) LoRA rank and projection layer ablations

paired with quantization and knowledge distillation for edge deployment; and (vi) evaluation extended to latency, perceptual quality (PESQ/STOI), and fairness across age/gender/dialect subgroups.

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